

ARSHDEEP SINGH

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SUMMARY

Final year undergraduate at IIT Roorkee with practical experience in backend systems and distributed infrastructure. Interested in databases, system design, cryptography, and web development. Passionate about open-source software, production engineering, and DevOps.

EDUCATION

Bachelor of Technology - Indian Institute of Technology, Roorkee • Roorkee

2023 - 2027

SKILLS

Programming Languages: C, Rust, Go, Solidity, Bash, JavaScript, TypeScript, C#

Tools and Technologies: Git, PostgreSQL, Hono, Drizzle, Redis, Linux, Docker, AWS, Kubernetes, NextJS

EXPERIENCE

Software Engineer, Tal Boss (by Grapevine) 🔗

May 2026 – August 2026, Bangalore, India

- Contributed to core job discovery and candidate matching services serving 10k+ users; integrated LinkedIn auth, profile extraction, and verification for candidate onboarding workflows; experimented with Tinker Labs models for matching optimization.
- Worked on candidate and job profile indexing in Qdrant; explored cloud storage migration from Azure Blob to AWS S3 for cost optimization and evaluated schema improvements for scaling backend infrastructure.
- Contributed to asynchronous job queue infrastructure using BullMQ, implemented multiple worker handlers for background processing, integrated observability stack (Signoz) for real-time debugging, and worked with GetStream for event handling and messaging.

Contributor, Protocol Labs Dev Guild 🔗

April 2025 - October 2025, Remote

- Contributing to open-source projects in LabOS as part of PLDG Cohort 3.
- Built Event Harvester - event aggregator for Luma and Cryptonomad with data cleaning and Airtable integration.
- Built Ask Racha - docs Q&A; parsed docs and repos, embedded into Qdrant (vector DB) for low-latency retrieval with a lightweight retrieval API.

LFX Mentee, WasmEdge 🔗

September 2025 - November 2025, Remote

- Implemented core features for the cross-platform wasmedgeup installer: remove, versioning, and plugin management.
- Added hardware/system detection for optimal runtime configuration (CPU features, GPU backends: CUDA/ROCm/OpenCL) with validation and tests.
- Strengthened reliability with checksum verification, documentation, unit tests, and a CI workflow across Linux, macOS, and Windows.

PROJECT

Watchdog 🔗 | Rust

December 2024 - February 2026

- Rust-based server access management tool that tracks admin attempts (sudo/su) via SSH, allows/denies based on public key, logs to Slack.
- It provides easy granting/revoking access to 4 servers to 50+ team members of SDS Labs through pull requests on a key house repository.
- Designed and implemented a project-as-group architecture by mapping projects to Linux groups for access control.

VortexDB 🔗 | Rust, HNSW

October 2025 - Present

- Lightweight vector database built from scratch in Rust with pluggable indexers (Flat, KD-Tree, HNSW for approximate nearest neighbor search).
- Multiple distance metrics (Euclidean, Manhattan, Hamming, Cosine); pluggable storage engine (RocksDB for persistence, in-memory ephemeral) behind trait.
- Axum HTTP API, gRPC server, Ratatui TUI, Python SDK; flexible similarity computation and client integration.

agentq 🔗 | Rust

September 2025 - Present

- Embeddable job queue for agent tool calls with idempotency keys caching outputs; retries return cache instead of re-running, preventing duplicate writes and charges.
- Priority lanes (High, Low) with independent concurrency limits for different budgets; backpressure prevents unbounded queue growth and protects against rate limits and metered services.
- Published on crates.io with no external service or Redis dependency; embedded directly in binary with failure isolation (errors don't cascade), cancel-safe semantics, and a minimal API footprint designed for production agent frameworks.

LEADERSHIP & VOLUNTEERING

Developer, SDS Labs 🔗

February 2024 - Present, Roorkee, India

- Involved in maintaining and developing applications for campus students, as well as those open to the internet.
- Regularly conducted hackathons (1000+ participants), public lectures, and competitions to foster the technical culture on the campus.
- Competed in 6+ hackathons and CTF events under SDS Labs team.

RECOMMENDATIONS

Jesper Pedersen

Google Summer of Code Administrator • PostgreSQL • jesperpedersen.db@gmail.com